

User-centered design and assistive technology

As user-centered design means trying to solve problems that people commonly experience rather than trying to solve only those problems that you experience personally. You've also been learning that good user-centered design incorporates the assistive technology, or AT, that many people use to interact with their devices. AT works in things such as color modification, voice control, switch devices, and screen readers.

While assistive technologies are often created to support people with disabilities, they're also used by many people who don't have disabilities or who don't identify as having disabilities. For example, consider the technology that allows someone to pinch a smartphone screen to expand an image or piece of text. That technology is used both by people who have visual impairments and by people who don't.

But when exactly do UX designers incorporate assistive technologies into their designs? The answer is straightforward: Effective UX designers consider assistive technology throughout the design process, from understanding users' experiences, to identifying users' needs, to designing and testing solutions.

Color modification

First, consider assistive technologies for color modification. A UX designer is creating a job-application form that users can access and complete on their mobile devices. During the early phases of the design process, the designer tries to understand how users might experience the form on a variety of devices. The designer discovers that some users use both light-mode settings and dark-mode settings to comfortably see and use their devices, switching back and forth depending on their surroundings and the application they're using.

Based on this information, the designer plans to include a "light mode/dark mode" toggle in the application form itself so users can easily select how they want to view the application form. This feature minimizes the risk that certain users will miss key instructions or interactive features, like text-input fields and buttons.

Switch devices

Now consider an example where a team has to redesign a product after forgetting to account for switch devices. (Remember that a switch device is a technology that replaces a keyboard or a mouse. Switch devices enable people to interact with devices without using one or both of those pieces of hardware.)

A UX design team is working on a new website for a local restaurant. The team finishes its first prototype, which allows test users to simulate navigating the live website. One user reports that they can't use their mouse to navigate the prototype. Keyboard navigation is an answer to that problem: It's a common switch technology that enables users to navigate websites and apps strictly with their keyboards, so they don't have to scroll and click with a mouse.

The principles of user-centered design demand that websites support keyboard navigation. And in many places, laws and regulations require the same! This means the design team must return to an earlier phase of its design process and re-work its prototype to support that switch device. The team could have avoided this re-work if it had used the early phases of its process to better understand how its users use switch devices.

Voice control

Now consider voice control technologies, which enable people to interact with products by using their voices rather than mouses, keyboards, or touch screens. For example, a UX designer is creating a vocal user interface, or VUI, for a smart thermostat. The VUI will enable users to interact with the thermostat by speaking.

At the very beginning of their process, the designer researches how users interact with similar voice control technologies: The designer observes people interacting with other smart thermostats to get a sense for how different users express the same commands. With this information, the designer creates a VUI that enables people to use multiple questions or commands to achieve the same tasks, like lowering the temperature or turning the thermostat off and on.

Screen readers

Finally, consider user-centered design and screen readers. Screen readers read the text on a screen. They also read alternative text, or alt text. Alt text is text that describes non-textual elements of a screen, such as images and graphics. Screen readers read alt text aloud so users can understand the screen's non-textual elements and how they fit with the screen's textual content.

When designing with screen readers in mind, it's crucial to consider your users, determine which images they'll find meaningful (rather than merely decorative), and then create clear alt text to describe those visuals. With clear alt text in place, users can get value out of your product's images even if they have visual impairments or low-bandwidth connections that make it difficult for images to load.

Grow through daily observations

If you haven't already, start observing how you use assistive technologies to interact with everyday products. Do you ever turn on closed captions while streaming a movie? Ask a smart speaker to play your favorite song? Or use the tab key to move from one text-input field to another? If so, you've used assistive technology. The next time, stop to consider how the designer incorporated that technology into their design process. Over time, these small observations can help you grow into the user-centered designer you want to be!

Defining Tactile User Needs for Accessible Design

Effective tactile UX designers begin the design thinking process by understanding their project's goals, constraints, and business requirements. They also prioritize the "empathize" phase, connecting with users, gathering insights, and developing detailed tactile user personas.

Process Flow for Tactile Design Thinking:

1. Empathize,
2. **Define Tactile Needs**,
3. Ideate,
4. Prototype, and
5. Test.

After empathizing, designers move to the "define" phase, focusing on the key question: "Which of my users' tactile needs or challenges are most critical for my design to address?" They use a combination of tools to answer this.

Tactile User Stories

While a tactile user persona represents a group of users with similar tactile needs, tactile user stories provide a more focused perspective. A tactile user story is a one-sentence narrative from a persona's viewpoint, detailing who they are, what they want to do with a tactile product, and why. The following template can be used:

As a [type of user with tactile needs], I want to [action with tactile product] so that [benefit related to tactile experience].

For example, considering a user who would use a tactile drawing tool like Graffiti:

"As a visually impaired student, I want to create tactile drawings of diagrams and graphs so that I can understand and learn complex concepts independently."

Tactile User Journeys

After creating tactile personas and stories, designers map out each persona's tactile user journey. This is a series of experiences the user has while trying to achieve their goal using a tactile product. It might be their experience with existing tactile tools, similar products, or their experience in the absence of such tools.

For example, a tactile user journey for a visually impaired artist might include:

- **Current Journey:** Relying on verbal descriptions or assistance to understand visual art.
- **Desired Journey:** Independently creating and interpreting tactile art using a tool that provides clear tactile feedback.

Tactile Problem Statements

With detailed tactile user journeys, designers identify the problems their tactile design must solve. They distill these problems into tactile problem statements.

A tactile problem statement summarizes who the user is, what they need from a tactile design, and why. The following template can be used:

[Name of tactile user persona] is a [type of user with tactile needs] who needs [type of tactile user experience] because [benefits of tactile user experience].

For example, a tactile problem statement for a user needing a tactile drawing tool:

"Alex, a visually impaired student, needs a tactile drawing tool that allows for precise and easily interpretable raised lines, so that they can independently create and understand visual information in their studies."

Effective tactile UX designers create problem statements for each tactile user persona. This ensures that the design addresses the needs of various users with diverse tactile requirements.

Key considerations for Tactile needs:

- **Material properties:** Consider texture, durability, and safety of materials.
- **Tactile feedback:** How will the user know they are interacting correctly?
- **Spatial awareness:** How will the user navigate and understand the tactile information?
- **Accessibility standards:** Adhere to accessibility guidelines for tactile products.
- **Durability and Cleanliness:** Consider the longevity of the product, and how it can be kept clean.

Tactile Ideation: Generating Solutions for Accessible Experiences

In the second phase of design thinking, designers use their empathy research to define and prioritize the user problems they need to address, especially for users with visual impairments. In the third phase, they **ideate tactile solutions** to these problems.

Process Flow for Tactile Design Thinking: 1. Empathize, 2. Define, 3. **Tactile Ideate**, 4. Prototype, and 5. Test.

The **tactile ideation** phase is about fostering creativity and brainstorming a wide range of potential tactile solutions. Designers often collaborate with team members, accessibility experts, and potential users with visual impairments. The goal is to generate innovative and unconventional ideas that enhance accessibility through touch.

Early in the ideation process, it's crucial to suspend assumptions and preconceptions. The focus is on generating as many tactile alternatives as possible without immediate evaluation. Evaluation and selection come later, allowing for a thorough comparison of options.

Here are some common ideation techniques, adapted for tactile product design:

"How Might We?" for Tactile Experiences

Designers take the problem statements related to tactile accessibility and reframe them as "How Might We" (HMW) questions. This helps transform challenges into opportunities for tactile innovation. For example, regarding a tactile drawing tool like Graffiti:

- **Original Problem:** Users with visual impairments struggle to create and interpret complex tactile drawings.
- **HMW Questions:**
 - How might we create a tactile drawing surface that allows for precise and easily interpretable raised lines?
 - How might we develop a stylus that provides clear tactile feedback during drawing?
 - How might we incorporate different textures and patterns to represent various elements in a tactile drawing?
 - How might we create a tactile drawing that can be easily erased and reused?

After generating HMW questions, designers brainstorm numerous tactile solutions.

Tactile Sketching and Material Exploration

Instead of visual sketches, we focus on **tactile sketches** and **material exploration**.

- **Rapid Material Sampling:** Designers gather diverse materials with varying textures, thicknesses, and densities. They quickly create small tactile samples to represent potential features.
- **Tactile Mock-ups:** Using materials like clay, foam, or textured paper, designers create quick physical mock-ups of tactile elements, such as raised lines, dots, or shapes.
- **Texture Mapping:** Creating small grids with different textures to represent different values, or line types.

Tactile Competitive Audits

Designers analyze existing tactile products, focusing on their tactile features and accessibility.

- **Tactile Evaluation:** Designers assess the quality and clarity of tactile elements in competitor products.
- **Accessibility Analysis:** They evaluate how well competitor products meet the needs of users with visual impairments.
- **Material Comparison:** They compare the materials used in competitor products and identify potential alternatives.

SCAMPER for Tactile Design

SCAMPER is adapted to focus on tactile modifications:

- **Substitute:** Substitute materials with different textures or densities.
- **Combine:** Combine multiple tactile elements into a single feature.
- **Adapt:** Adapt existing tactile features to new contexts.
- **Modify:** Modify the size, shape, or texture of tactile elements.
- **Put to Another Use:** Explore alternative uses for existing tactile materials or technologies.
- **Eliminate:** Eliminate unnecessary tactile elements.
- **Rearrange:** Rearrange the layout of tactile elements for improved usability.

Tactile Collaboration and User Involvement

Collaboration with users with visual impairments is essential.

- **Tactile Co-creation Workshops:** Designers conduct workshops where users with visual impairments contribute to the ideation process.
- **Accessibility Feedback Sessions:** Designers gather feedback on potential tactile solutions from accessibility experts and users.
- **Material Testing Panels:** Users test different materials and provide feedback on their tactile properties.

Business Requirements and Tactile Feasibility

Designers consider the feasibility of tactile solutions within project constraints.

- **Material Cost Analysis:** They evaluate the cost of materials for tactile prototypes and final products.
- **Manufacturing Feasibility:** They assess the feasibility of manufacturing tactile elements with the available resources.
- **Durability and Maintenance:** They consider the durability and maintenance requirements of tactile materials.

Prototyping Tactile Solutions for Enhanced Accessibility

When a designer deeply understands their users, clearly defines their needs, and generates innovative ideas, it's time to move into the fourth phase of design thinking: **prototyping tactile solutions**.

Process Flow for Tactile Design Thinking: 1. Empathize, 2. Define, 3. Ideate, 4. **Tactile Prototype**, and 5. Test.

In this phase, the goal is not to produce a polished final product, but rather to construct tangible, early models—tactile prototypes—that allow designers to experience how the solution will feel and function for users, especially those with visual impairments. These prototypes also serve as powerful tools for communicating the design's potential to stakeholders and potential users. How, then, do designers prototype for tactile experiences?

Tactile Information Architecture and Spatial Mapping

For tactile products like those inspired by Graphiti, the organization of tactile elements is crucial. Users must be able to navigate and interpret the information through touch. This is where **tactile information architecture** and **spatial mapping** come into play.

Instead of a traditional sitemap, we might create a **tactile map** or a **spatial layout plan**. Imagine a grid-based system where each square represents a tactile element or a specific area of interaction.

- **Tactile Hierarchy:** This focuses on how different textures, heights, or patterns are used to indicate importance or category. For example, raised dots might represent primary elements, while textured lines indicate secondary information.
- **Tactile Sequence:** This defines the order in which users explore the tactile information. For example, a raised arrow might guide the user from one section to the next.

Tactile Wireframing and Material Exploration

Instead of visual wireframes, we create **tactile wireframes** or **material samples**. These are physical representations of the design, focusing on the texture, shape, and arrangement of tactile features.

- **Paper-Based Tactile Wireframes:** We might use embossed paper, textured cardstock, or layered materials to create initial tactile layouts.
- **Material Exploration Boards:** Designers can create boards with various materials to test different textures, thicknesses, and densities. This helps in selecting the most suitable materials for the final product. Examples could include different types of plastics, resins, or textured papers.
- **3D Printed Tactile Wireframes:** 3D printing can be used to rapidly produce tactile elements with precise dimensions and textures.

Low-Fidelity (Lo-Fi) Tactile Prototypes

Lo-fi tactile prototypes are simple, interactive models that provide a basic understanding of the product's tactile functionality. This could involve:

- Creating a grid system with raised dots or lines to represent a drawing surface.
- Using Velcro or magnets to attach and rearrange tactile elements.
- Creating simple embossed or thermoformed sheets to test different textures.

High-Fidelity (Hi-Fi) Tactile Prototypes

Hi-fi tactile prototypes closely resemble the final product, incorporating all the intended tactile features, materials, and interactivity.

- Using precision embossing or thermoforming to create detailed tactile graphics.
- Integrating electronic components to provide audio feedback or haptic vibrations.
- Creating a fully functional drawing surface with a stylus that creates raised lines.

Iterative Design and User Feedback

Throughout the prototyping process, designers must continuously seek feedback from users, especially those with visual impairments. This involves:

- Conducting tactile evaluations to assess the clarity and effectiveness of the tactile information.
- Gathering feedback on the comfort, durability, and usability of the materials.
- Observing how users interact with the prototypes and identifying any areas for improvement.

Prototyping and Tactile Testing

Prototyping and testing are deeply intertwined in tactile design. Each prototype iteration provides an opportunity to gather valuable feedback and refine the design.

Key Takeaways for Tactile Prototyping:

- Emphasize tactile information architecture and spatial mapping.
- Explore a wide range of materials and textures.
- Create both lo-fi and hi-fi tactile prototypes.
- Prioritize user feedback and iterative design.
- Use 3D printing, Embossing, and thermoforming to create accurate tactile representations.

By focusing on these principles, designers can create tactile products that are not only functional but also empowering and accessible for all users.

Testing Tactile Solutions for Enhanced Accessibility

Once designers have created a tactile prototype, it's essential to test it rigorously. Testing is the fifth phase of the tactile design thinking process, and it often occurs in iterative cycles with prototyping.

Process Flow for Tactile Design Thinking: 1. Empathize, 2. Define, 3. Ideate, 4. Prototype, and 5. **Test Tactile Solutions.**

Consider a designer creating a tactile drawing tool similar to Graphiti. They've developed a prototype with raised lines and textured surfaces. Now, they must test its effectiveness.

Effective tactile testing aims to understand how users perceive, interpret, and interact with the tactile features. To ensure reliable and unbiased results, a well-structured testing plan is crucial. Let's explore the key elements of tactile UX testing:

Who will participate in the tests?

- Early tests might involve colleagues, accessibility experts, or individuals familiar with tactile materials.
- Later tests should include real-life users who align with the target user personas, particularly those with visual impairments or other tactile sensitivities.

What types of tests will be conducted?

- **Tactile Perception Tests:** Assess the clarity and distinctiveness of different textures and raised elements.
- **Usability Tests:** Observe how users navigate and interact with the tactile interface.
- **Task-Based Evaluations:** Evaluate users' ability to complete specific tasks, such as creating a tactile drawing or identifying tactile patterns.
- **Comparative Analysis:** Compare the prototype to existing tactile products or alternative solutions.
- **Error Rate Analysis:** Track how often a user makes a mistake when interacting with the product.

When and where will the tests be conducted?

- Tests can be conducted in controlled environments, such as accessibility labs, or in real-world settings.
- Consider the impact of ambient noise and lighting on tactile perception.
- Virtual testing can be done through video calls, with users describing their tactile explorations.

Why are the tests conducted?

- Validate the effectiveness of the tactile design in conveying information.
- Identify any usability issues or areas for improvement.
- Ensure the design meets accessibility standards and guidelines.

- To discover if the product is enjoyable and empowering to use.

How will participants engage with the prototype?

- Provide clear instructions and allow ample time for exploration.
- Offer alternative methods of interaction, such as verbal descriptions or assistive technologies.
- Ensure the testing environment is accessible and comfortable for all participants.
- Consider using gloves, or other tools to simulate different tactile sensitivities.

How will information be collected?

- Observe users' interactions and note any difficulties or hesitations.
- Conduct interviews and gather feedback on their experiences.
- Use questionnaires or surveys to collect quantitative data.
- Record the time it takes to complete tasks.
- Record user's verbal descriptions of the tactile object.

Example: Testing a Tactile Drawing Tool

For a tactile drawing tool, the designer might:

- Ask visually impaired participants to create specific drawings and evaluate the clarity of the raised lines.
- Observe how users navigate the drawing surface and identify any areas of confusion.
- Gather feedback on the comfort and usability of the stylus.
- Test the durability of the raised lines, and how easy they are to erase.